

Team 3

CAMPUS WAYFINDING CHALLENGES:

A Navigation System for Hanyang ERICA

Human-Computer System Design

Problem Definition

1. Building Identification Issues

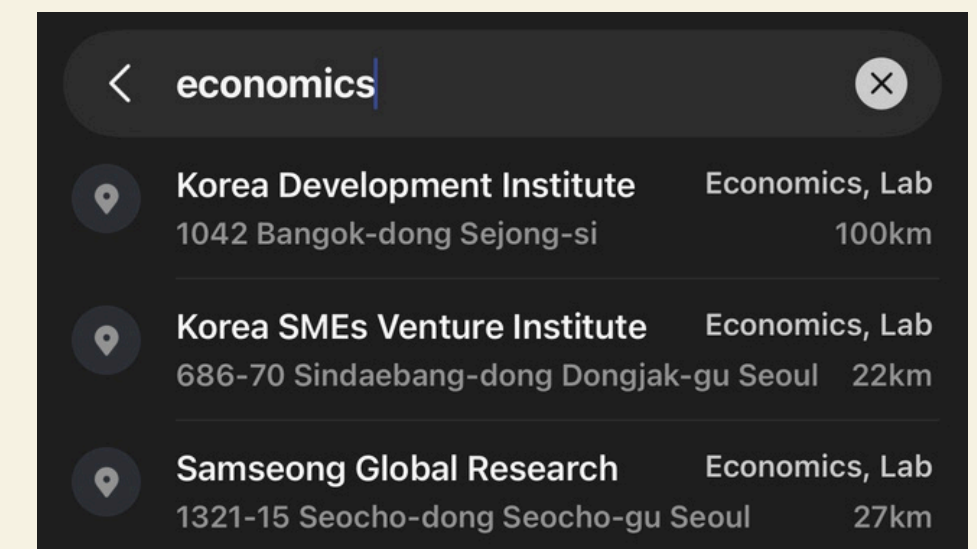
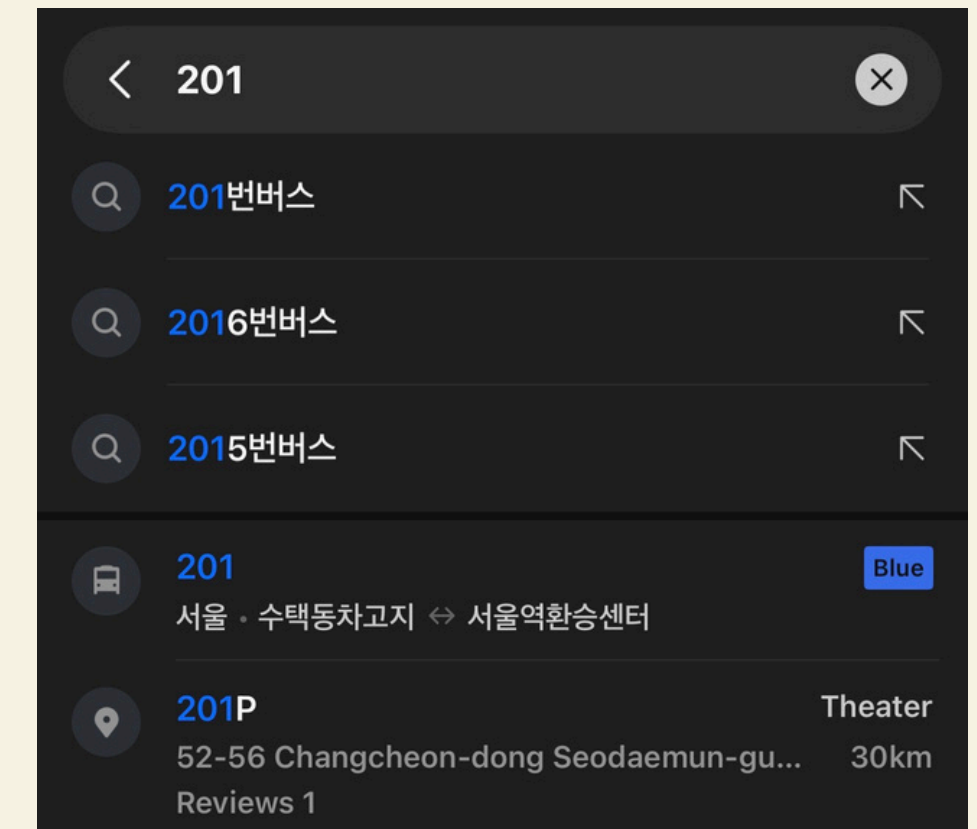
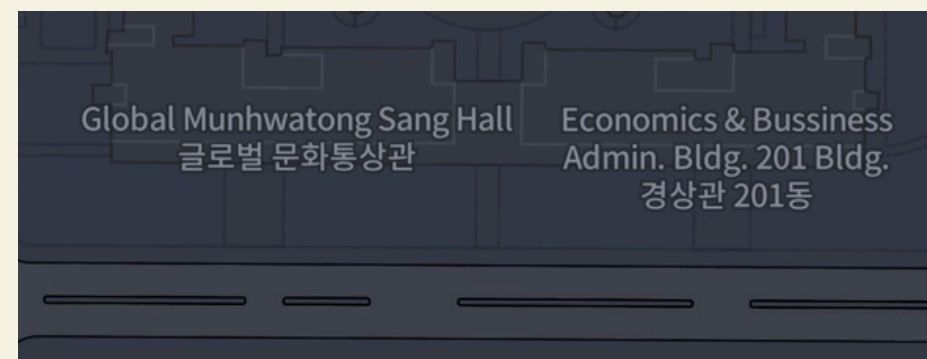
Class timetables only list building numbers and room numbers, which are confusing.

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2. Language and Search Limitations

In Naver, though buildings are labeled in both Korean and English, they cannot be searched easily.



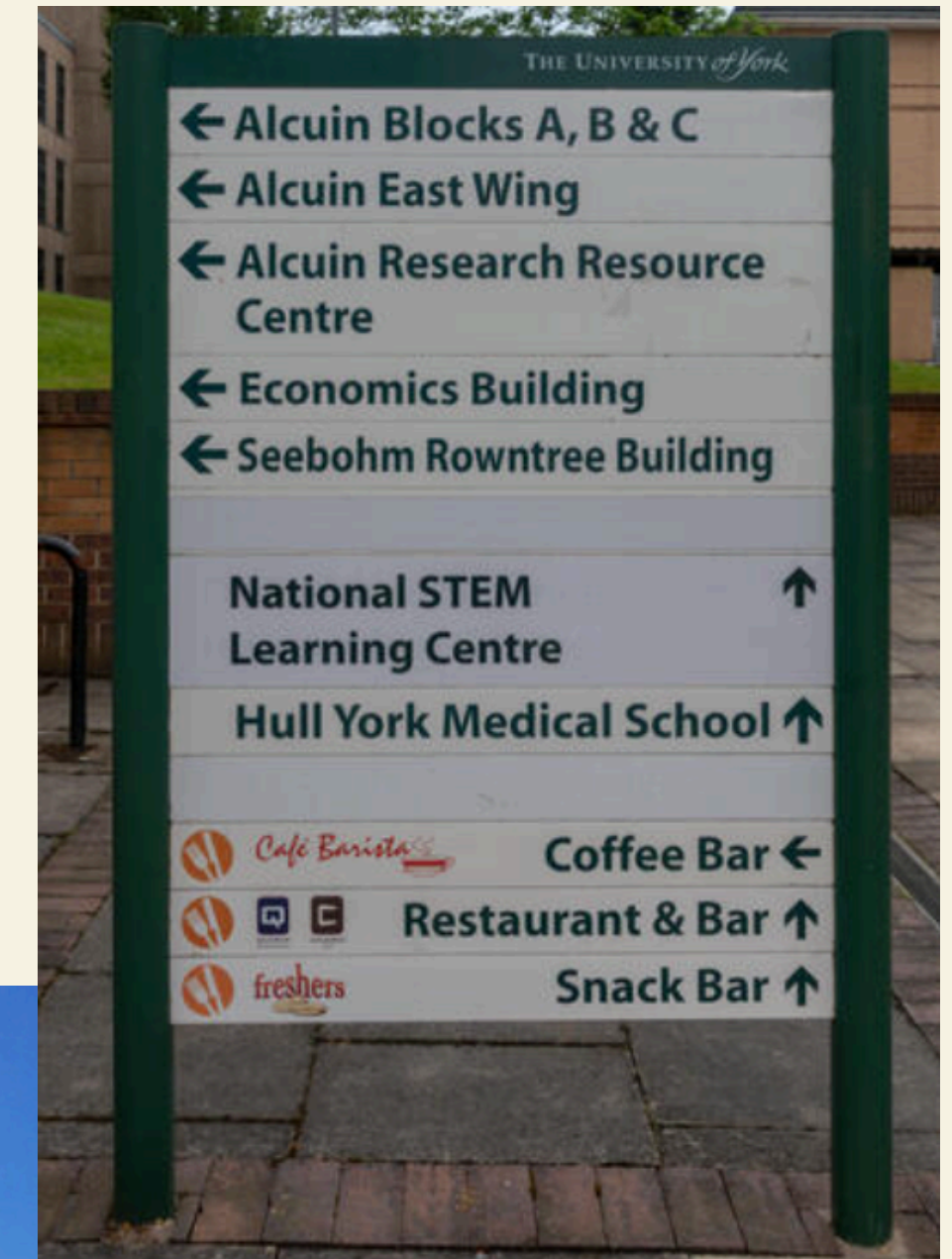
3. Difficulty Finding Classrooms

Navigation apps typically guide users only to the building, not to the specific classroom.

Large buildings can be confusing to navigate without floor maps or guidance.

4. Insufficient Campus Signage

Unclear signage causes difficulty in confirming building numbers and directions.



Improvement Requirements

1. Multilingual Search Functionality

- Allow searches using English names, Korean names, or building numbers.
- Display dual-language search results so users can easily compare them with campus signage.

2. Building Number and Map Integration

- Integrate users' timetables and provide classroom-level navigation.
- Integrate campus shuttle bus schedules and highlight important campus facilities and services.

3. Indoor Navigation & Wayfinding

- Provide simple navigation and easy search functions.
- Offer step-by-step directions both across campus.

4. Visual Building Identification

- Provide building photos, clear building numbers, and map labels.
- Use landmark-based navigation cues.

Questionnaire Study

We conducted a questionnaire survey with 45 participants to understand students' experiences navigating the ERICA campus.

The questionnaire consisted of three sections:

1. Language & Campus Familiarity
2. Campus Navigation Experience
3. Navigation Tools Experience

Participant Characteristics

Participants were grouped by Korean language familiarity:

1. Native Korean speakers (n = 15)
2. Foreign students with Korean familiarity (n = 15)
3. Foreign students with no Korean familiarity (n = 15)

Total participants: 45

Hypotheses

Hypothesis 1: Language Familiarity and Navigation Difficulty

H1 (Alternative Hypothesis):
Korean language familiarity significantly affects difficulty locating buildings on campus.

Hypothesis 2: Indoor Navigation Difficulty

H1 (Alternative Hypothesis):
Students experience significant difficulty locating classrooms or facilities inside buildings.

Hypothesis 3: Navigation Tool Helpfulness

H1 (Alternative Hypothesis):
There is a significant difference in perceived helpfulness of navigation tools between language groups.

Data Analysis Methods

1. Descriptive Statistics

- Mean values for Likert scale responses
- Frequencies and percentages for categorical responses

2. Inferential Statistics

i. Chi-square test

- Used to examine relationships between categorical variables.

ii. Binomial Test

- To compare an observed proportion against a specified expected proportion.

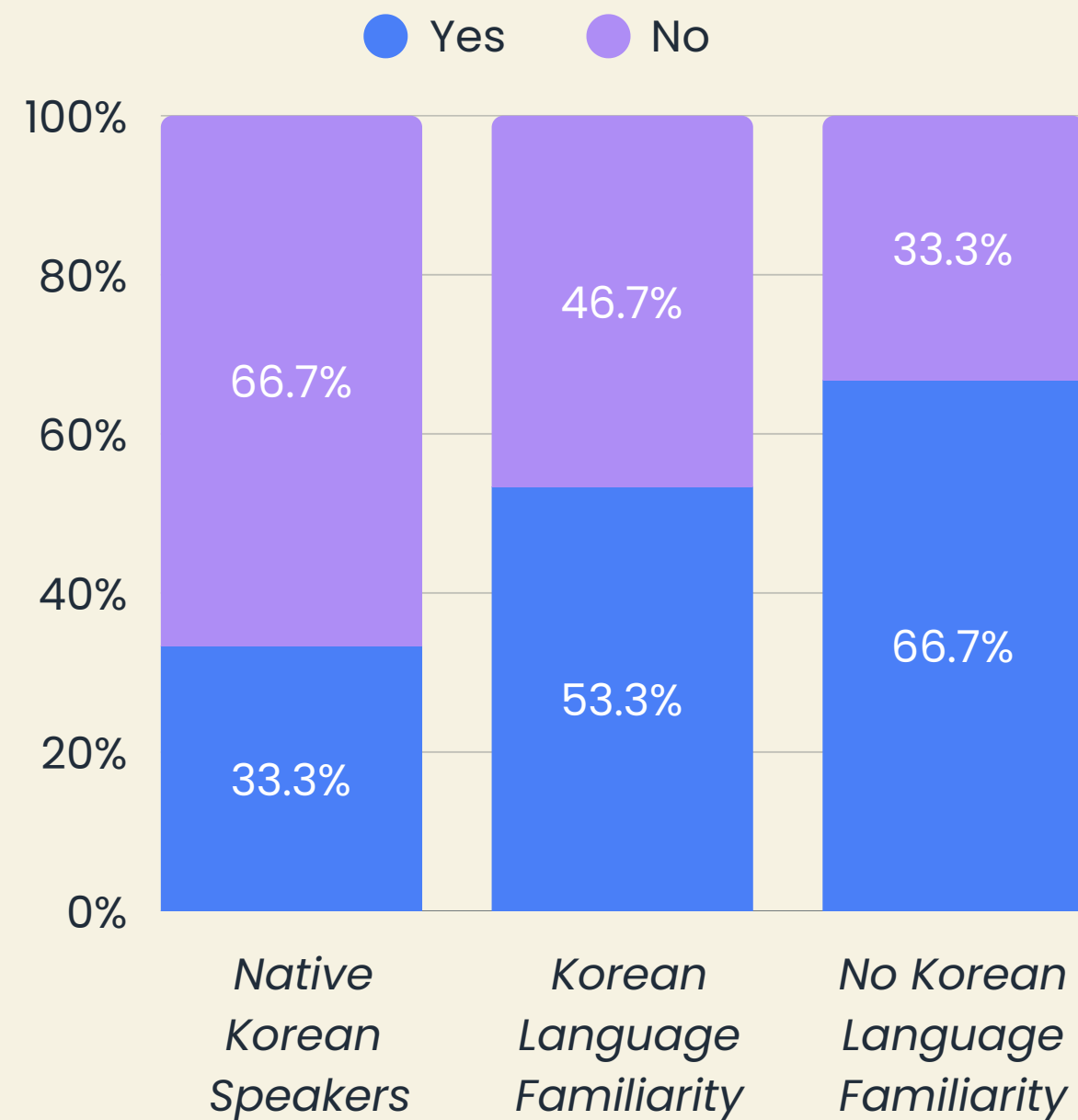
iii. One-way ANOVA

- Used to compare mean values across the three language groups

1. Chi-Square Test of Independence

H1 (Alternative Hypothesis):

Korean language familiarity significantly affects difficulty locating buildings on campus.



$$\chi^2 (2, N=45) = 3.379$$

$$p = 0.185$$

Findings

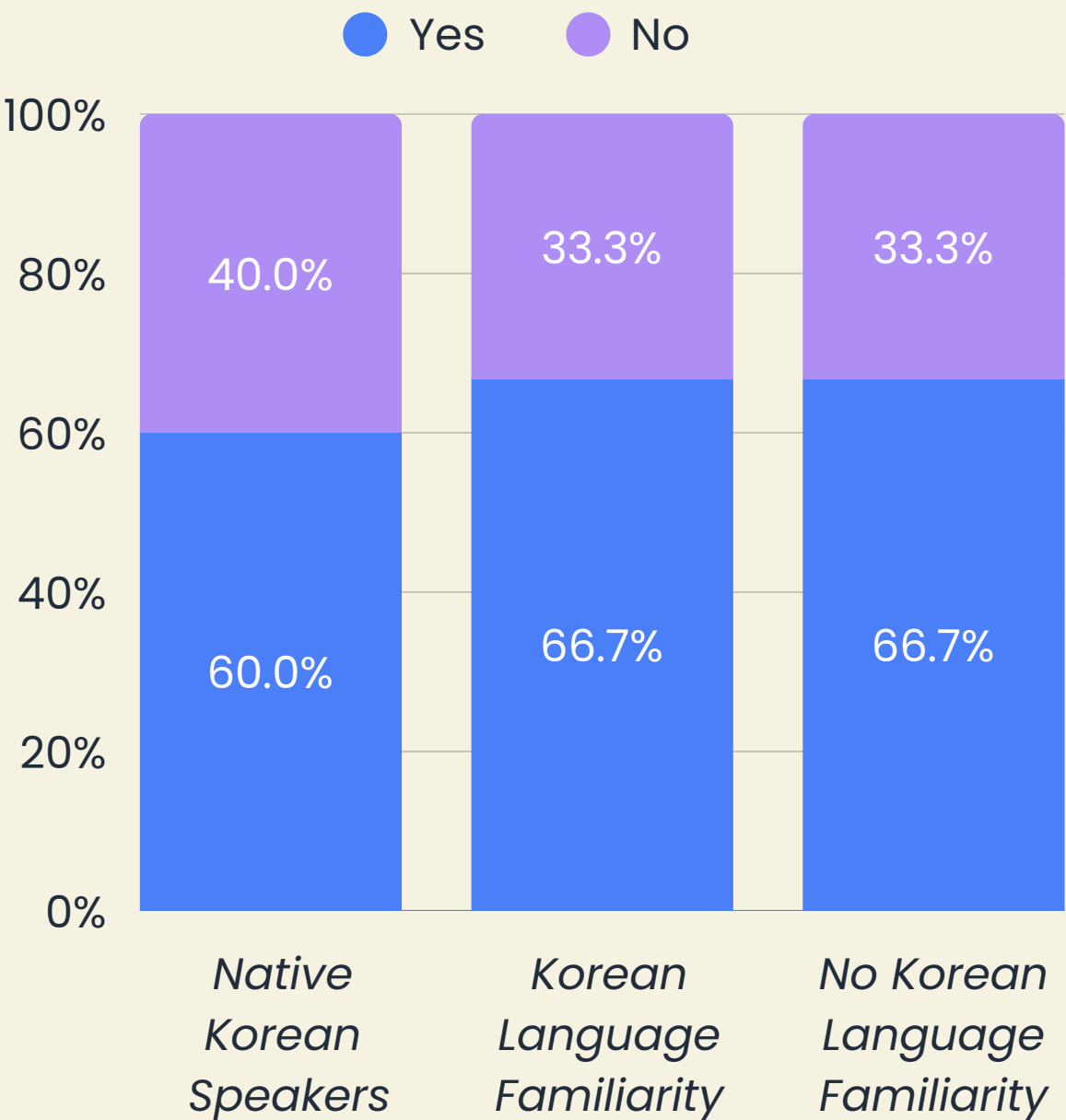
No statistically significant relationship was found between Korean language familiarity and difficulty locating buildings.

However, descriptive results showed a trend where students with lower Korean familiarity experienced greater navigation difficulty.

2. Binomial Test

H1 (Alternative Hypothesis):

Students experience significant difficulty locating classrooms or facilities inside buildings.



Test Proportion: 0.5 (50%)

One-Tailed $p = 0.036$

Observed Proportion: 0.64 (64%)

Findings

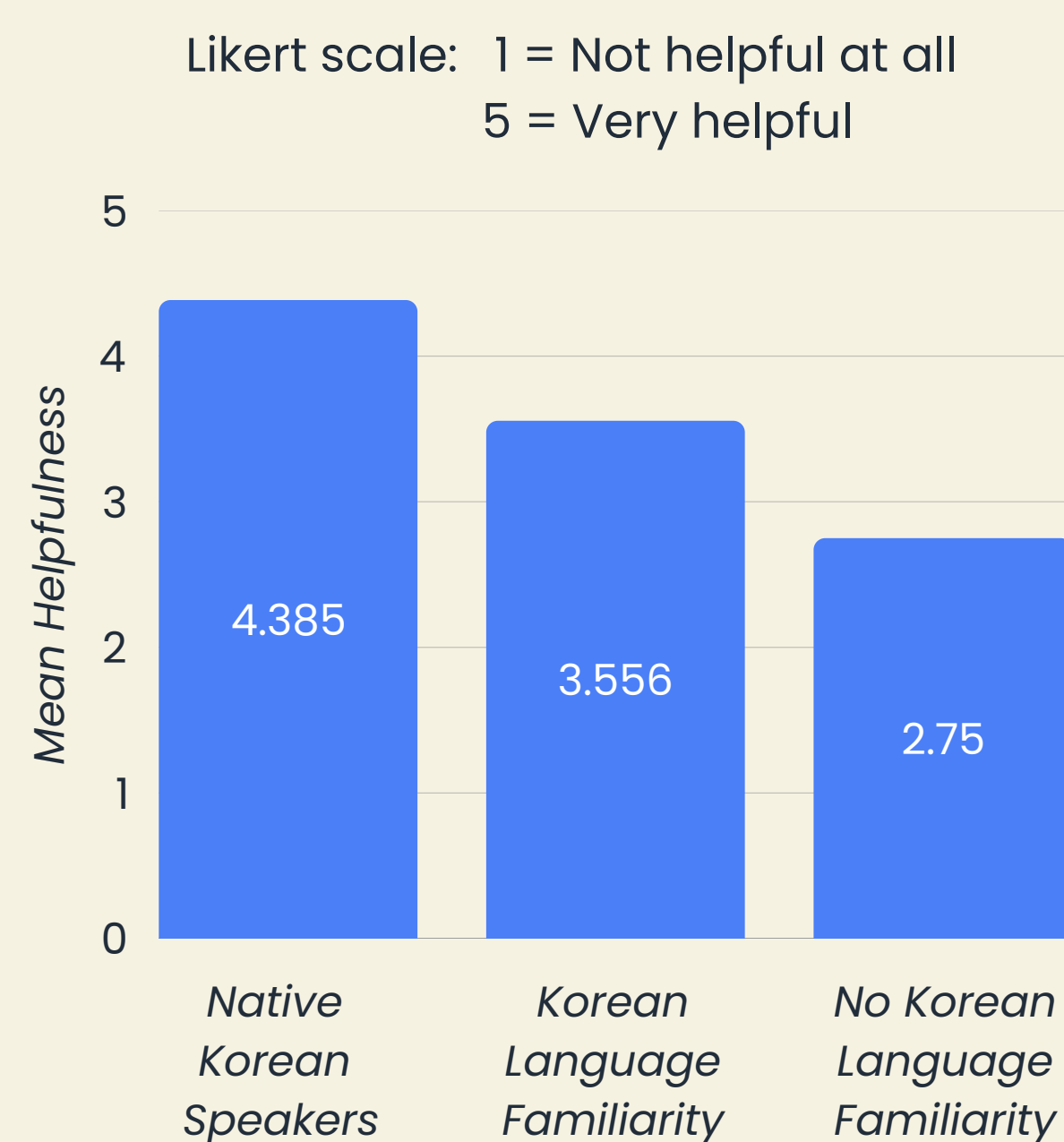
The result was statistically significant.

More than half of the students experienced difficulty locating classrooms or facilities inside buildings.

3. One-Way ANOVA (Scheffe Post-Hoc)

H1 (Alternative Hypothesis):

There is a significant difference in perceived helpfulness of **Naver Map** between language groups.



Native Korean speakers vs Korean familiarity

p-value

0.028 (Yes)

Native Korean speakers vs No Korean familiarity

0.001 (Yes)

Korean familiarity vs No Korean familiarity

0.151 (No)

Post Hoc Findings

Native Korean speakers rated Naver Map significantly **more helpful** than:

- Korean-familiar students
- Students with no Korean familiarity

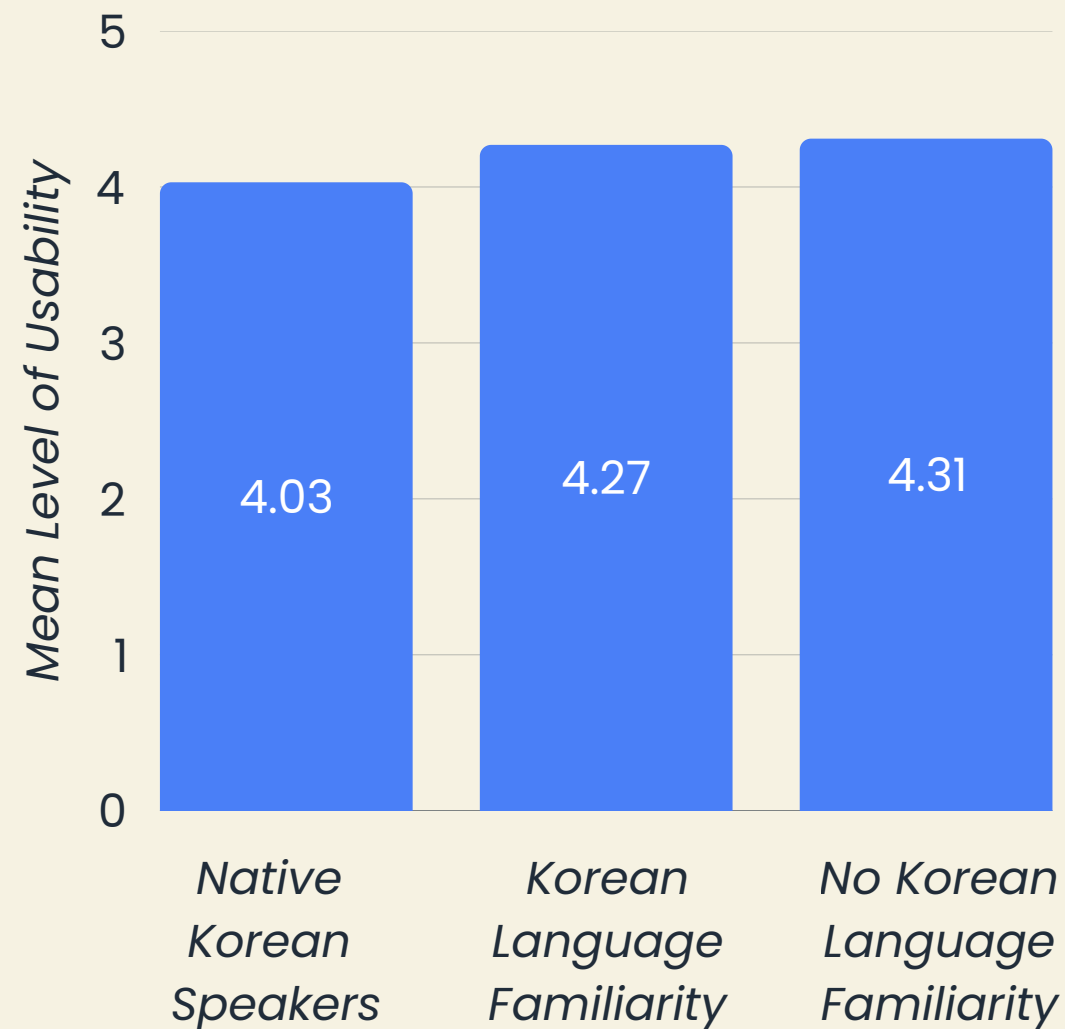
No significant difference was found between Korean-familiar students and students with no Korean familiarity.

Evaluation Results

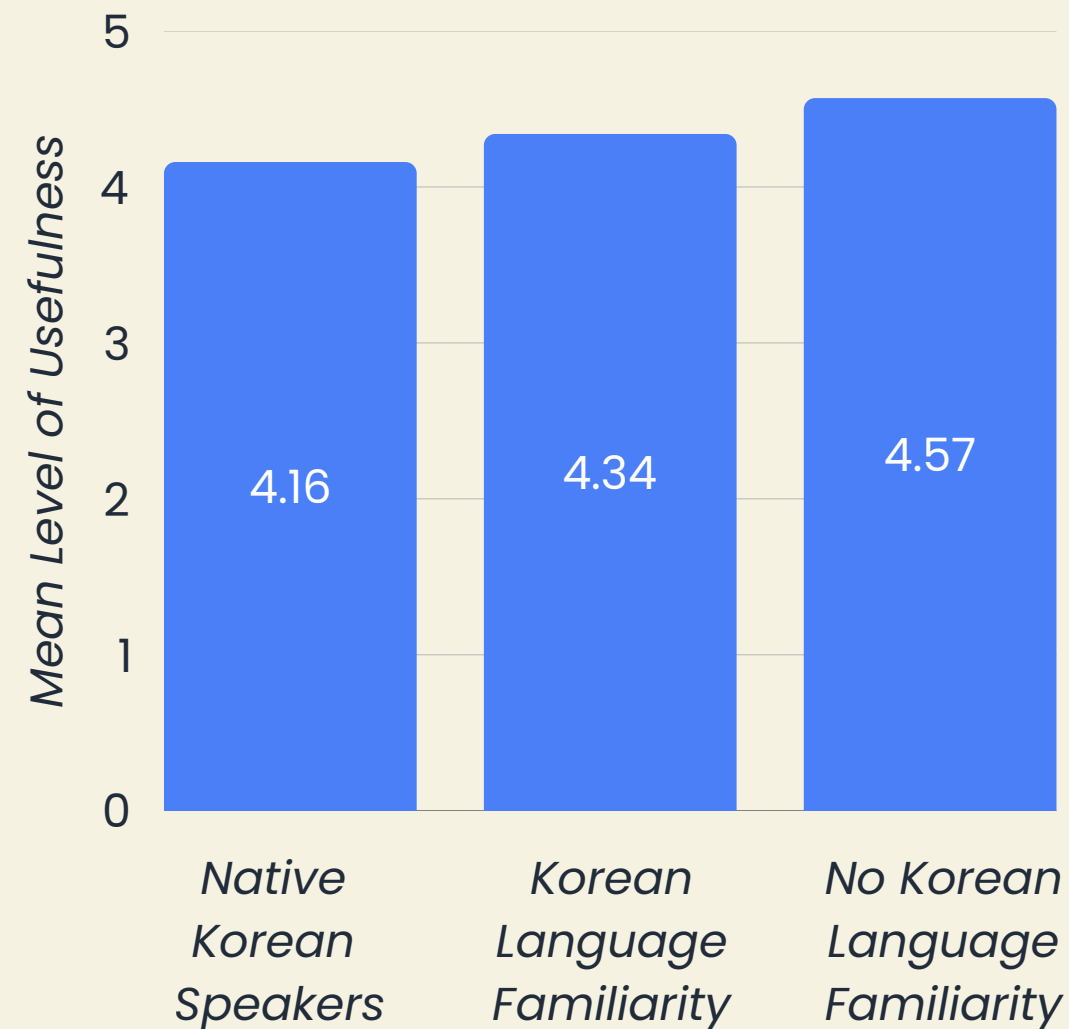
Sample Size: n = 10 per group

Likert scale: 1 = Not helpful at all
5 = Very helpful

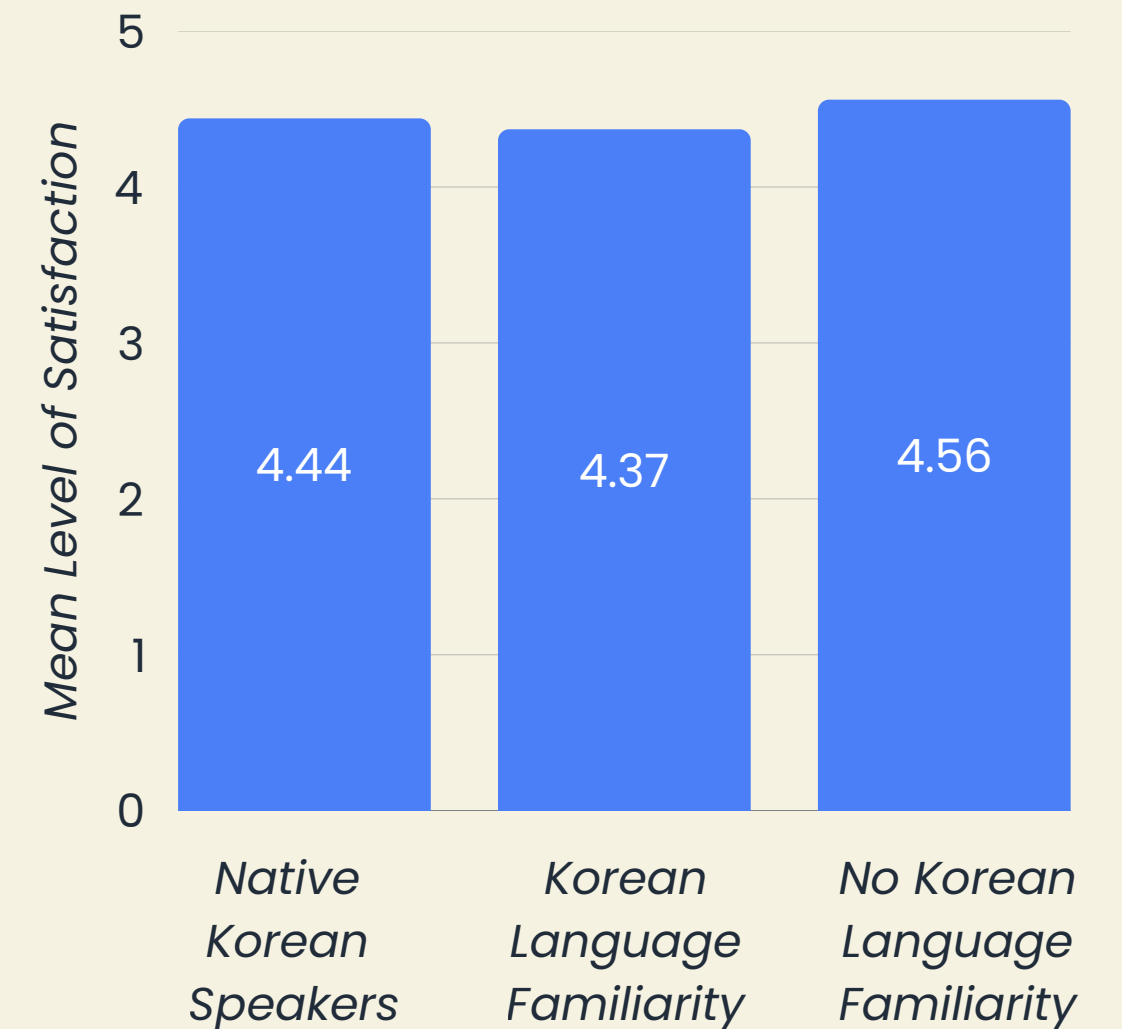
Prototype Usability



Prototype Usefulness



User Satisfaction



Study Limitations

1. Small Sample Size

The data might not statistically strong enough due to number of participants.

3. Prior familiarity with applications

Some participants may be more experienced with one navigation app than another, affecting their ratings.

2. Self-reported Data

Helpfulness ratings are based on participants' perceptions, which can be subjective and influenced by personal preferences

4. No Comparison Baseline

The prototype was rated on its own, not tested against Naver Map or the official campus map.

Major Findings

1. Prototype Rated Highly by Every Group

Participants rated the prototype above 4 out of 5 for usability, **usefulness and satisfaction in all three language groups.**

3. Indoor Navigation Is the Biggest Gap

64% of students had difficulty finding a classroom inside a building, **significantly more than half ($p = 0.036$).**

2. Language Familiarity Affects Map App Experience

Native Korean speakers rated Naver Map significantly more helpful than **international students, with or without Korean ability ($p = 0.028$ and 0.001).**

Future Plans

1. Indoor Navigation Analysis

Future work will focus on analyzing indoor navigation challenges and developing indoor-level navigation features within the app.

2. 3D Map Exploration

Although this study uses a 2D map design, future research will examine whether 3D maps improve wayfinding performance.

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References

Yvette, C. C., & Song, Y. (2026). *A bilingual app for campus wayfinding and local cultural immersion*.

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Klippel, A., Hirtle, S., & Davies, C. (2010). *You-are-here maps: Creating spatial awareness through map-like representations*. *Spatial Cognition & Computation*, 10(2–3), 83–93.

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